Chapter 21

Volatility Modeling: Local, Stochastic, Jump, and Rough †

Section 19.4 named three families of model that resolve the contradiction between a formula with one volatility and a market that quotes many, and said what each one buys and what it costs. That is enough to tell them apart; it is not enough to choose one and defend the choice. This chapter develops how each is built, how it is calibrated, and what its particular failure costs, and adds a fourth, rough volatility, which is the most recent development and only became usable after about 2015. The mathematics is deferred to [TN21]; what stays here is what each model captures, where it fails, and how one would tell.

The organizing question is not "which model is right." No model here is right. The question is which model is wrong in a way that does not matter for the trade in front of you, and answering it requires knowing precisely how each one fails.

The order is logical rather than historical (Section 21.13 gives the history). Local volatility comes first, in Section 21.2, because an exact fit to today's surface forces it on you, and Section 21.3 shows what the fit costs in dynamics; Heston's stochastic volatility, Merton's jumps and their combination in Bates follow. Section 21.7 measures all three against one fact, the $T^{-0.4}$ power law the index skew follows, and none of them reproduces it. That fact produced the fourth family; the later sections turn to the hybrid models desks run, to calibration, and to what the disagreement between models costs.

One convention holds for every worked example in the chapter: $S_0 = 100$ and $r = q = 0$. The forward then equals the spot, "at the money" means a strike of 100 at every maturity, and the numbers measure the models rather than the carry. The equations keep r and q, because the models are used with them; the examples drop them, because they would only obscure the comparison.

21.1 WHAT A VOLATILITY MODEL MUST DO

Two requirements compete, and almost every model satisfies one at the expense of the other.

Fit today's surface. A model that does not reproduce the quoted vanilla prices is arbitrageable against the market by anyone who trades the vanillas. This is a hard constraint, not a preference.

Get the dynamics right. The model must also say how the surface will move, because the value of anything path-dependent depends on that. Section 19.11 showed the point in miniature: the delta of a plain call already depends on whether the smile is sticky in strike or in moneyness, and that is a question about dynamics, not about today's prices.

Nothing in the vanilla market pins down the dynamics. Today's surface is a snapshot of the risk-neutral distribution at each maturity; infinitely many processes produce the same set of terminal distributions and behave completely differently in between. That degree of freedom is the subject of this chapter, and Section 21.11 measures what it is worth in money.

Table 21.1 states the trade each of the four families makes, and it is worth reading before

the sections that develop them: no row buys both requirements, which is why the chapter ends in a choice rather than in a winner.

Table 21.1: What each family buys and what it costs.

Family	Buys	Costs
Local volatility	Fits any arbitrage-free surface exactly; stays complete	Dynamics wrong: the forward smile (quoted on a future date) flattens; the market's does not
Stochastic volatility	Realistic dynamics, stationary forward smile	Cannot fit the short end; market incomplete
Jumps	Short-dated skew a diffusion cannot produce	Skew dies too fast with maturity; unhedgeable by construction
Rough volatility	The observed skew term structure from one parameter	Not Markovian; simulation is expensive

21.2 LOCAL VOLATILITY

Let volatility be a deterministic function of price and time:

$$dS = (r - q)\,S\,dt + \sigma_{\text{loc}}(S, t)\,S\,dW \tag{21.1}$$

Dupire's result is that for any arbitrage-free surface of call prices $C(K, T)$ there is exactly one function σ_{loc} in (21.1) reproducing every one of them, and it can be read off directly, evaluated at $S = K$, $t = T$:

$$\sigma_{\text{loc}}^2(K, T) = \frac{\dfrac{\partial C}{\partial T} + (r - q)K\dfrac{\partial C}{\partial K} + qC}{\frac{1}{2}K^2\dfrac{\partial^2 C}{\partial K^2}} \tag{21.2}$$

Three things about (21.2) are easy to read past.

The denominator is the implied density. It is $\frac{1}{2}K^2e^{-rT}$ times the Breeden–Litzenberger density of (19.1). So local volatility is a ratio of two things the surface already contains, and it exists precisely when the density is positive and the numerator is not negative, which are Section 19.5's butterfly and calendar conditions again, now as an existence condition for a model.

The model is complete. There is one Brownian motion and one traded asset, so Section 13.4's replication argument survives intact and every option is hedgeable with the underlying. That is a real advantage: a local-volatility book has a delta that means something.

The fit is exact by construction, not by calibration. There are no parameters to search over. This is why local volatility remains the baseline for any payoff fixed by today's surface alone, such as an unquoted European.

The simplest local volatility is a single power of the price, the constant elasticity of variance model of Cox (1975), $\sigma_{\text{loc}}(S) = \sigma S^{\beta-1}$ with σ a scale constant: with $\beta < 1$ volatility rises as the price falls, a skew from one parameter, and $\beta = 1$ and $\beta = 0$ are the lognormal and normal cases. It fits no particular surface, but its exponent survives as SABR's β (Section 29.9).

Example 21.1 Local volatility is about twice as steep as implied

Take the surface Heston's model (Section 21.4) gives at the base case of Figure 21.2 (any arbitrage-free surface will do) and apply (21.2). Its call prices come from Heston's semi-analytic formula, and the strike derivatives are central differences in log-strike $y = \log K$ (with $C_y = \partial C/\partial y$, $K^2\,\partial^2 C/\partial K^2 = C_{yy} - C_y$), which keeps the wings numerically stable, and $\partial C/\partial T$ is a central

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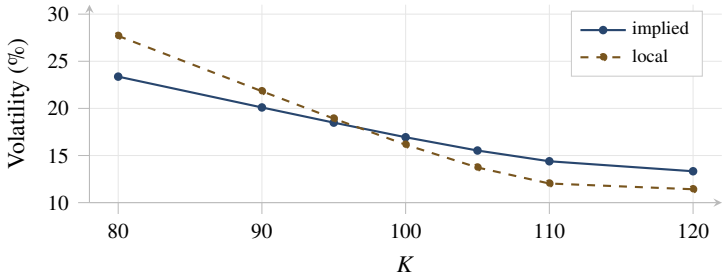


Figure 21.1: Local and implied volatility at one year, from the same Heston surface.

difference in maturity. These are formula prices, not quotes, so none of [TN21]'s differencing noise arises here. Both functions are read at the same argument K, but it means different things on the two sides: for σ_{impl} it is the strike of an option expiring in a year, and for σ_{loc} it is the price level at which that volatility is in force, a year from now. With the chapter's convention, $S_0 = 100$ and $r = q = 0$:

Strike	Implied vol	Local vol
80	23.37%	27.69%
90	20.10%	21.82%
95	18.49%	18.92%
100	16.94%	16.15%
105	15.53%	13.73%
110	14.39%	12.03%
120	13.33%	11.42%

Measured as slopes in $k = \log(K/S_0)$ at the money, across the ±5% pair (the convention (21.3) fixes), implied volatility falls at -0.296 per unit of k (about 3 volatility points from 95 to 105) and local volatility at -0.518, a ratio of **1.75**.

Figure 21.1 shows the two curves together. This is the "rule of two," and the reason for it is an averaging argument. An implied volatility is the single number that reproduces the price of an option struck at K, so it is an average of the local volatilities the price path visits on its way to K. Averaging flattens. To produce a given slope in the average, the underlying function must slope about twice as steeply.

So a steep local-volatility surface, however alarming its wings look, is not a calibration failure; it is arithmetic.

21.3 WHAT LOCAL VOLATILITY GETS WRONG

Section 19.4 said local volatility fits today's prices exactly and forecasts their evolution badly. Here is the mechanism and its size.

Because σ_{loc} is a fixed function of S and t, the model has no freedom left once it is calibrated. In particular it has a definite prediction about the **forward smile**: the smile that will be quoted at some future date, for options starting then. And the prediction is wrong in a specific direction.

One measurement convention is needed first, because from here on the chapter reports skews as numbers rather than shapes. Section 19.1 used "skew" for the shape: implied volatility falling as the strike rises. Measured, it is the slope of implied volatility in **log-moneyness**, taken at the money, and the tables below report the second derivative beside it:

$$\text{skew}(T) = \left.\frac{\partial \sigma_{\text{impl}}}{\partial k}\right|_{k=0}, \qquad \text{curvature}(T) = \left.\frac{\partial^2 \sigma_{\text{impl}}}{\partial k^2}\right|_{k=0} \tag{21.3}$$

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with $k = \log(K/F)$, where F is the forward price to the option's maturity, $F = S_0 e^{(r-q)T}$ for an asset paying a continuous dividend yield q. Under the chapter's convention $r = q = 0$, so $F = S_0$ throughout and "at the money" and "at the forward" name the same strike. **Curvature** is the smile's convexity in the same variable: zero for a straight line, and positive when both wings sit above the tangent at the money. Both are estimated in practice by finite differences across a pair of strikes either side of the forward (and the forward itself, for the curvature): placed symmetrically in k, at Fe^{-h} and Fe^{h}, where a table states a bump h, and at 95 and 105 (the "±5% strike pair") where it does not. At so narrow a spacing the two placements barely differ.

Three properties are worth fixing now, because every table in this chapter reports them. *Log-moneyness rather than strike* makes the number scale-free, so two underlyings trading at 40 and 4,000 report the same skew for the same shape. *The sign is negative on an equity surface*, and a steeper skew is a more negative number. *The value depends on the width of the strike pair*, because the smile's slope changes away from the money: the one-year base-case smile of Section 21.4 measures −0.296 across the ±5% strike pair and −0.282 at $h = 0.10$. Each table states its own bump, and skews are comparable down a column rather than across tables.

Example 21.2 The forward smile flattens

Take Example 21.1's surface and measure both skews ±5% around the forward, the local one at the date equal to each maturity:

Maturity	Implied skew	Local-vol skew
3 months	−0.479	−0.981
6 months	−0.403	−0.770
1 year	−0.296	−0.518
2 years	−0.196	−0.320
3 years	−0.148	−0.231

The local volatility function's own skew decays by a factor of more than four between three months and three years. A one-year option starting in three years is priced from the local volatilities between years three and four, whose skew starts at −0.231, where today's one-year option is priced from skews of −0.98 to −0.52; so the model predicts that the three-year-forward smile will be far flatter than today's spot smile.

The market does not agree. Forward-starting options (options whose strike is set at the money on a future date) and cliquets, chains of them that reset one after another, trade with a skew comparable to today's: the smile is roughly *stationary* in moneyness, not decaying. So a local-volatility model marks anything that gains when the future skew steepens too cheap (an out-of-the-money forward-starting put, a forward-starting risk reversal that is long the put and short the call) and its mirror image too expensive, and, worse, hedges both with the wrong vega profile.

Two products with identical vanilla hedges can have quite different forward-smile exposure, and local volatility, holding one flattened forward smile, cannot tell them apart.

There is a second, subtler consequence. Because local volatility ties σ to S, a fall in the spot mechanically raises volatility along the calibrated function. That makes the model's delta systematically different from the Black–Scholes delta, in a direction Section 19.11's vocabulary can name: a rise in the spot *raises* a fixed strike's implied volatility under sticky delta and *lowers* it under local volatility, so the local-volatility delta sits below even the sticky-strike (Black–Scholes) delta. The measured index behavior of Section 19.11 lies between the two: below Black–Scholes, but not as far below as local volatility puts it.

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21.4 STOCHASTIC VOLATILITY

Let volatility be a second random process, correlated with the price:

$$dS = (r-q)S\,dt + \sqrt{v}\,S\,dW_1, \qquad dv = \kappa(\theta - v)\,dt + \sigma_v\sqrt{v}\,dW_2, \qquad dW_1\,dW_2 = \rho\,dt \quad (21.4)$$

The variance v mean-reverts at rate κ toward a long-run level θ, with volatility-of-volatility σ_v, and it is correlated with the price at ρ. The square root in the diffusion keeps v nonnegative, though it can touch zero unless $2\kappa\theta \geq \sigma_v^2$, the Feller condition, which fitted equity parameters usually violate (the base case used throughout the chapter, $v_0 = \theta = 0.04$, $\kappa = 1.5$, $\sigma_v = 0.6$, $\rho = -0.7$, has $2\kappa\theta = 0.12$ against $\sigma_v^2 = 0.36$); that matters for simulation, where a discretized variance can go negative, not for the formula. The square root is also what makes the model tractable: the characteristic function $\varphi(u) = \mathbb{E}\left[e^{iu\log S_T}\right]$ (risk-neutral, $i = \sqrt{-1}$) is available in closed form, and a call price follows from it by Fourier inversion rather than by simulation,

$$C = S_0 e^{-qT} P_1 - K e^{-rT} P_2, \qquad P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \mathrm{Re}\left[\frac{e^{-iu\log K}\varphi_j(u)}{iu}\right] du \quad (21.5)$$

where $\varphi_2 = \varphi$ and $\varphi_1(u) = \varphi(u - i)/\varphi(-i)$, which is the same function under the measure that takes the stock itself as numeraire; P_2 is the risk-neutral probability that the call finishes in the money and P_1 the same probability under that stock measure, Heston's $N(d_2)$ and $N(d_1)$. And φ is elementary. Writing $b = \kappa - \rho\sigma_v iu$ and

$$d = \sqrt{b^2 + \sigma_v^2\left(u^2 + iu\right)}, \qquad g = \frac{b-d}{b+d}$$

the characteristic function is

$$\varphi(u) = \exp\left\{iu\left(\log S_0 + (r-q)T\right) + A(T) + B(T)\,v_0\right\}$$

with

$$B(T) = \frac{b-d}{\sigma_v^2}\cdot\frac{1-e^{-dT}}{1-ge^{-dT}}, \qquad A(T) = \frac{\kappa\theta}{\sigma_v^2}\left[(b-d)T - 2\log\frac{1-ge^{-dT}}{1-g}\right]$$

That is the whole of it: a square root, an exponential and a logarithm in u, with no integral and no recursion anywhere. This is what "semi-closed form" means: no formula for the price, but none needed, because everything except a single one-dimensional integral is written above. It is also why a whole surface calibrates in seconds. One warning: d and the logarithm are complex, and the algebra pins only d^2, which either square root satisfies, and the two give algebraically equivalent formulas for A and B. Take the principal root, as written above, and the logarithm stays continuous; take the other and it jumps by $2\pi i$, destroying the pricing integral at long maturities. [TN21] derives all of it and calls that failure the "little Heston trap."

Heston is the standard model for three reasons that have little to do with realism. The semi-closed form is the first. It has five parameters (v_0, θ, κ, σ_v, ρ), which is enough to fit a smile and few enough to interpret. And every parameter maps onto a feature of the surface a trader already thinks in terms of.

Figure 21.2 shows the two parameters a trader adjusts most. Table 21.2 puts numbers on four of the five; the fifth, v_0, stays at 0.04 throughout. Its skew and curvature are first and second central differences of implied volatility in log-moneyness at $h = 0.10$.

Correlation produces the skew, and produces it for the right reason. At $\rho = 0$ the skew vanishes: the smile is symmetric to three decimals, which is Section 19.1's currency shape. Making ρ negative tilts it, and the economics is Section 19.9's leverage explanation written as a parameter: when volatility rises as the price falls, low strikes become expensive.

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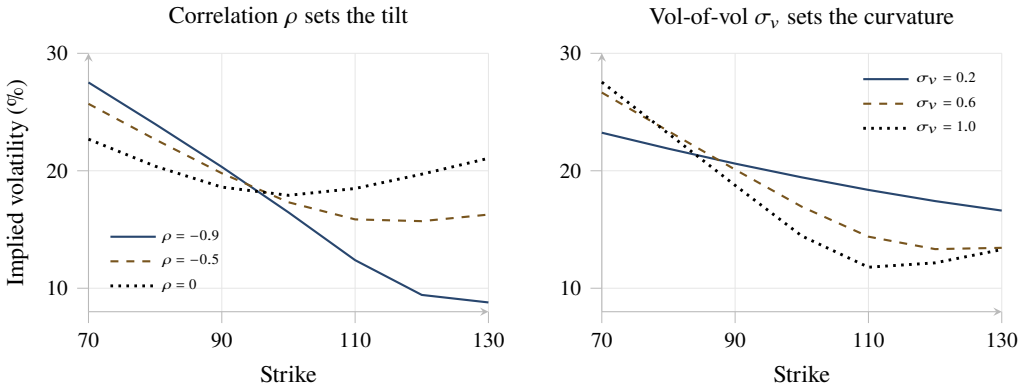


Figure 21.2: One-year Heston smiles. Base case $v_0 = \theta = 0.04$, $\kappa = 1.5$, $\sigma_v = 0.6$, $\rho = -0.7$; each panel varies one parameter.

Table 21.2: One-year at-the-money level, skew, and curvature as each parameter is varied from the base case.

	ATM (%)	Skew	Curvature
Base ($\rho = -0.7$, $\sigma_v = 0.6$, $\kappa = 1.5$, $\theta = 0.04$)	16.94	−0.282	0.36
$\rho = -0.9$	16.46	−0.398	−0.58
$\rho = 0$	17.91	0.000	1.27
$\sigma_v = 0.2$	19.44	−0.112	−0.01
$\sigma_v = 1.0$	14.45	−0.340	1.40
$\kappa = 5$	18.94	−0.155	0.03
$\theta = 0.09$	22.70	−0.227	0.08

Vol-of-vol produces the curvature. At $\sigma_v = 0.2$ the curvature is −0.01 and the smile is a straight line to two decimals; at $\sigma_v = 1.0$ it is 1.40 and both wings lift. This is the mechanism of Section 19.9: uncertainty about volatility fattens both tails of the terminal distribution.

The two interact, and that is the practical difficulty. Raising σ_v from 0.2 to 1.0 also steepens the skew from −0.112 to −0.340 and drops the at-the-money level from 19.44% to 14.45%. No parameter moves one feature alone, so calibration is a genuine optimization rather than a sequence of one-dimensional fits, and the fitted parameters are correlated enough that reporting them individually is misleading.

Mean reversion flattens everything. At $\kappa = 5$ the skew falls to −0.155 and the curvature to 0.03: fast mean reversion gives variance less time to wander, so the terminal distribution is closer to lognormal. This is the averaging that flattens the skew with maturity beyond Section 19.1's change of unit, and it is why κ is hard to identify from a single maturity's smile: its effect looks like a longer maturity.

Diffusive variance fails at the short end. Variance is a diffusion, so over a vanishingly short horizon it has not had time to move, and however the parameters are set the at-the-money skew approaches a finite limit as $T \to 0$ instead of steepening as the market's does. In the base case the one-month skew is 1.77 times the one-year skew (slopes at $h = 0.01$), where the market's power law gives about 2.7. Section 21.7 measures the gap; Sections 21.5 and 21.6 are the classical alternatives, jumps alone and jumps added to Heston.

Rate markets use SABR instead, built for refitting speed: its volatility is stochastic too, but an asymptotic expansion gives the implied volatility as a formula, so a whole grid of swaption smiles can be refitted many times a day (Section 29.9).

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21.5 JUMPS

Allow the price to move discontinuously. In Merton's model a diffusion is punctuated by jumps arriving at rate λ, each multiplying the price by a lognormal factor Y:

$$\frac{dS}{S} = (r - q - \lambda\bar{k})\,dt + \sigma\,dW + (Y-1)\,dN, \qquad \bar{k} = \mathbb{E}[Y] - 1 \tag{21.6}$$

where N is a Poisson process of intensity λ (a count of jumps that rises by one at random moments, λ times a year on average) and $\log Y$ is normal with mean $\mu_J = \log(1+\bar{k}) - \delta^2/2$ and variance δ^2. The $-\lambda\bar{k}$ in the drift is a compensator: it subtracts the jumps' expected contribution, so that jumps negative on average do not drag the price below the forward and $e^{-(r-q)t}S_t$ stays a martingale.

Conditional on exactly n jumps the terminal price is lognormal again, so the option is a Poisson-weighted sum of Black–Scholes–Merton values, which is what makes the model closed form:

$$C = \sum_{n=0}^{\infty} \frac{e^{-\lambda^* T}(\lambda^* T)^n}{n!}\, C_{\mathrm{BSM}}(\sigma_n, r_n), \qquad \lambda^* = \lambda(1+\bar{k})$$

with $\sigma_n^2 = \sigma^2 + n\delta^2/T$ and $r_n = r - \lambda\bar{k} + n\log(1+\bar{k})/T$. Each term is the price on the paths that happen to take n jumps; the intensity is λ^* rather than λ because the sum discounts at r while each term discounts at its own r_n.

Merton's jumps are few and large, about one every two years at the $\lambda = 0.5$ used below. The variance-gamma model of Madan, Carr, and Chang (1998) and its generalization by Carr, Geman, Madan, and Yor (2002), known as CGMY, go to the other extreme: infinitely many small jumps and typically no diffusion. They fit short-dated skew well and price by the same Fourier inversion as Heston, but their skew flattens with maturity just as Merton's does, and they are used mainly to fit and interpolate vanilla smiles.

The reason to reach for jumps is entirely about short maturities.

Example 21.3 What a jump does that a diffusion cannot

Take Merton's model with $\sigma = 17\%$, jump intensity $\lambda = 0.5$ a year, mean jump $\bar{k} = -10\%$, jump volatility $\delta = 13\%$:

Strike	1 month	3 months	1 year
85	30.99%	23.82%	21.21%
90	25.07%	21.78%	20.81%
95	20.57%	20.35%	20.47%
100	18.77%	19.49%	20.18%
105	18.25%	18.99%	19.94%
110	18.42%	18.73%	19.74%

At one month the smile falls from 31.0% at 85 to a low of 18.2% at 105 and turns up again beyond it, a range of 12.7 volatility points. At one year it falls all the way across the grid, from 21.2% at 85 to 19.7% at 110, a range of 1.5 points.

The smile's range collapses by a factor of more than eight as maturity goes from one month to one year. A diffusion cannot do this, because over a short horizon a continuous path simply cannot travel far enough to make a 15%-out-of-the-money put worth anything; only a discontinuity can.

That is also the criticism. Measured as the at-the-money skew (slopes at $h = 0.01$), the jump model's ratio between one month and one year is **3.7**; the power law index skews follow (Section 21.7) gives about **2.7**. Jumps overshoot the short end by more than a third, where a diffusion undershoots, and a model fitted to one-month options will misprice one-year options unless the jump parameters are allowed to vary with maturity—which is to say, unless the model is

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abandoned.

Section 19.4 noted that jumps break completeness for a sharper reason than stochastic volatility does: a jump of unknown size cannot be hedged by any position established before it. Stochastic volatility leaves a risk no position in the stock alone can hedge, but a second option hedges it, canceling the vega term of Chapter 18's expansion, equation (18.1), as Example 18.7 did; jump risk is not hedged by the stock and any finite set of options, because no such set pays off correctly for every possible jump size. A jump is a move too large for that expansion to describe, so canceling its terms, however completely, does not reach it. Section 13.4's argument said an incomplete market has no unique price; with jumps, it has no perfect hedge either: options can reduce the exposure, but no finite set of them removes it.

Who is paid for jump risk. Because the risk cannot be hedged away, someone must hold it, and they are paid to: part of Section 19.10's volatility risk premium compensates for diffusive variance a delta hedger cannot capture perfectly, and part for a discontinuity no hedge covers at all. The second part sits in the wings and at short maturities, where Example 21.3 put the jump model's contribution, which is convenient for calibration and treacherous for inference: a steep short-dated put wing is consistent both with a high real-world chance of jumps and with a large premium for bearing them, and option prices, which are risk-neutral, cannot separate the two.

For the desk, the consequence is a rule about who should be short. A delta-hedged book can hold diffusive variance risk and earn the premium on it. Jump risk does not diversify across stocks, because jumps cluster across underlyings (Chapter 23 shows the same clustering intraday), so a book that has systematically sold short-dated out-of-the-money options holds a correlated tail, and Chapter 22's case study of February 5, 2018 shows a crowd of short-volatility positions meeting such a tail.

21.6 STOCHASTIC VOLATILITY WITH JUMPS

One mechanism undershoots the short end and the other overshoots it, so the obvious move is to run both at once. Bates (1996) does exactly that: Heston's variance process, with Merton's jumps added to the price:

$$\frac{dS}{S} = (r - q - \lambda\bar{k})\,dt + \sqrt{v}\,dW_1 + (Y-1)\,dN, \qquad dv = \kappa(\theta - v)\,dt + \sigma_v\sqrt{v}\,dW_2 \quad (21.7)$$

with $dW_1\,dW_2 = \rho\,dt$, and N, Y and $\bar{k}$ as in (21.6). The jumps are taken independent of both Brownian motions.

That independence is what makes the model usable: $\log S_T$ is a sum of independent pieces, so its characteristic function is a product:

$$\varphi_{\text{Bates}}(u) = \varphi_{\text{Heston}}(u) \cdot \exp\left\{\lambda T\left(e^{\,iu\mu_J - u^2\delta^2/2} - 1\right) - iu\lambda\bar{k}T\right\}, \qquad \mu_J = \log(1+\bar{k}) - \frac{\delta^2}{2}$$

with φ_{Heston} exactly as Section 21.4 writes it. The price follows from the same inversion (21.5), subject to the same warning about the branch of d. One integral still prices it, which is why Bates is widely used for short-dated skew; [TN21] derives the product and recovers Merton's series from it.

Three parameters are added to Heston's five, and they act where Heston cannot reach:

- λ, the arrival rate, sets how much of the smile is made by jumps rather than by the diffusion.
- $\bar{k}$, the mean jump, tilts the short-dated smile, and a negative value is what makes short-dated puts expensive.

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- δ, the jump volatility, spreads the short-dated wings.

Example 21.4 shows the division of labor in the smiles themselves.

Example 21.4 Where each mechanism acts

Take Heston's base case with its diffusive variance cut to $v_0 = \theta = 0.025$ and Example 21.3's jumps added ($\lambda = 0.5$, mean jump -10%, jump volatility 13%), so that the two mechanisms together carry the base case's variance rate of 0.04 a year:

Strike	1 month	3 months	1 year
85	31.57%	26.08%	21.67%
90	26.52%	23.58%	20.36%
95	21.67%	20.68%	19.00%
100	17.17%	17.33%	17.55%
105	13.74%	14.08%	16.03%
110	15.06%	13.21%	14.57%

At one month the smile falls from 31.6% at 85 to 13.7% at 105, a range of 17.8 volatility points, and turns up again beyond it. Over the same strikes Heston's own one-month smile spans 11.1 points and the jump model's 12.7: the combination is steeper than either component, because the jump wing now sits on a diffusion that is itself skewed. At one year the range is 7.1 points, against Heston's 7.3 and the jump model's 1.5.

At one month the shape of the smile (the put wing and the upturn beyond 105) is the jump model's; at one year the smile is Heston's, tilt and all. Each mechanism carries the maturities the other cannot reach.

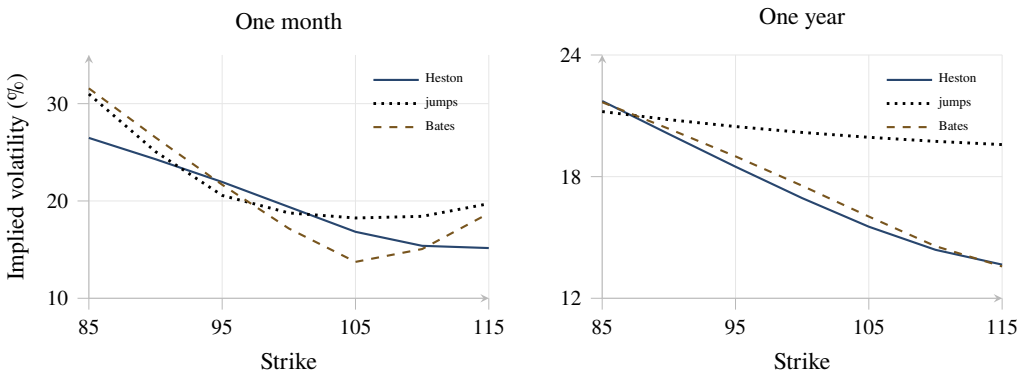


Figure 21.3: One-month and one-year smiles under Heston, Merton's jumps and Bates, with the parameters of Examples 21.3 and 21.4.

Figure 21.3 draws the comparison. In the left panel the Bates curve tracks the jump curve down the put wing and turns up with it past 105, sitting below it from the money upward because its diffusive variance is smaller, 0.025 against the jump model's $0.17^2 = 0.029$; in the right panel it lies on top of Heston's, and the jump curve is nearly flat. Table 21.3 puts numbers on what each of the three new parameters does to that picture. Every row resets $v_0 = \theta$ to hold the variance rate at the base case's; skew and curvature are central differences at $h = 0.10$, and the last column divides the one-month by the one-year slope at the money ($h = 0.01$, as in Figure 21.4), so it is not the quotient of the skew columns.

Read the one-year columns first. Down every row the skew stays between -0.24 and -0.36 and the curvature stays near zero, so a year out the three jump parameters are close to invisible and the smile is the one Table 21.2 described. At one month they are everything.

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Table 21.3: One-month and one-year at-the-money level, skew, and curvature as each jump parameter is varied from the Bates base case ($\lambda = 0.5$, $\bar{k} = -0.10$, $\delta = 0.13$).

		One month			One year			
	$v_0 = \theta$	ATM (%)	Skew	Curv.	ATM (%)	Skew	Curv.	**Ratio**
Base	0.0250	17.17	−0.531	7.11	17.55	−0.289	−0.44	2.92
$\lambda = 0.25$	0.0325	18.35	−0.495	3.42	17.23	−0.287	−0.02	2.23
$\lambda = 1.0$	0.0101	13.96	−0.584	17.02	18.19	−0.287	−1.28	6.20
$\bar{k} = -0.05$	0.0297	18.07	−0.380	6.34	17.35	−0.253	0.17	2.55
$\bar{k} = -0.15$	0.0169	15.19	−0.703	10.33	17.75	−0.346	−1.36	3.63
$\delta = 0.05$	0.0330	18.92	−0.538	1.29	17.49	−0.246	0.01	2.77
$\delta = 0.20$	0.0121	13.13	−0.400	19.15	16.64	−0.359	−0.13	3.18

The intensity sets how much of the smile the jumps make. Doubling λ to 1.0 leaves only a quarter of the variance in the diffusion, and the one-month at-the-money level falls from 17.2% to 14.0% while the curvature rises from 7.1 to 17.0: variance that used to sit under the whole distribution now sits in its tails. The one-year row barely notices.

The mean jump sets the tilt. From $\bar{k} = -0.05$ to -0.15 the one-month skew steepens from −0.380 to −0.703, and the one-month/one-year ratio goes from 2.55 to 3.63. This is the parameter that makes short-dated puts expensive, and it is the one a calibration to the put wing pins down first.

The jump volatility sets the wings. From $\delta = 0.05$ to 0.20 the one-month curvature goes from 1.3 to 19.2, and at the top of that range the skew flattens to −0.400, because a wide jump distribution fattens both tails rather than the left one alone.

What the combination buys is the last column. The base case's one-month slope at the money is **2.92** times its one-year slope, against the market's 2.70, where Heston alone reaches 1.77 and jumps alone 3.70; Section 21.7 draws all four against maturity. Mixing two errors of opposite sign cancels most of the short-end miss.

The long end is worse, not better. At three years the skew is 0.372 of its one-year value, against the market's 0.644, almost exactly the jump model's 0.367, and further off than Heston's own 0.494. Moving variance out of the diffusion and into jumps does not lift the term structure; it rotates it: jump skew is concentrated at short maturities and is negligible at long ones, so the short-end repair is paid for at the long end. Lowering the intensity trades one end against the other (at $\lambda = 0.25$ the three-year ratio improves to 0.42 while the one-month falls to 2.23, below the market); raising it misses both ends by more (6.20 and 0.317 at $\lambda = 1.0$).

Eight parameters are enough to put the fit where a particular book needs it, which is why many equity desks use some version of it for short-dated vanillas. They do not supply the power law the market's skew follows across maturities (Section 21.7): two unrelated mechanisms, each fitted to a different part of the surface, are a patch rather than an explanation.

21.7 THE SKEW TERM STRUCTURE, AND WHY IT BROKE ALL THREE MODELS

Sections 21.4, 21.5 and 21.6 each made a claim about how the skew behaves as maturity shortens. Put the three side by side against the market and the claims resolve into a single, sharp, measurable failure, the one the next section finally explains.

Write $\psi(T)$ for the *magnitude* of the at-the-money skew (21.3),

$$\psi(T) = -\left.\frac{\partial \sigma_{\text{impl}}}{\partial k}\right|_{k=0}$$

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which is positive on an equity surface, so that a steepening skew is a rising ψ. Ask how it behaves as $T \to 0$.

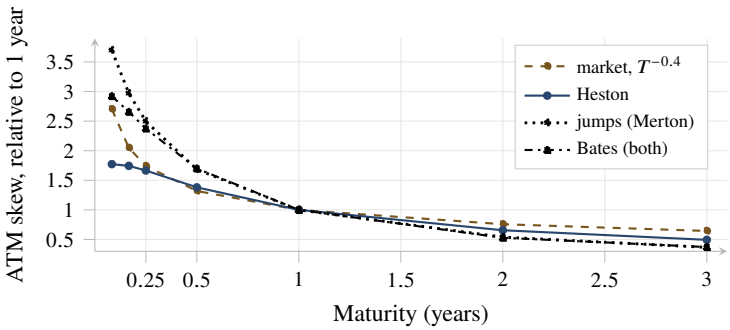


Figure 21.4: At-the-money skew against maturity, normalized to one year: Heston's base case, Example 21.3's jumps, Example 21.4's Bates, and the power law $T^{-0.4}$; slopes at the money, $h = 0.01$.

The empirical answer is a power law. Across equity indices, $\psi(T)$ is roughly proportional to $T^{-\alpha}$ with α around 0.4, at least from about a month to a few years: the skew keeps steepening as maturity shortens. Whether it goes on steepening below a month is disputed, and Section 21.8 returns to it.

A classical stochastic-volatility model cannot produce that. Section 21.4 said why: variance is a diffusion, so $\psi(T)$ approaches a finite constant as $T \to 0$ no matter how the parameters are set. In Figure 21.4 the Heston skew rises only 1.77 times from one year to one month, where the market rises 2.70 times.

At the long end the failure reverses. Heston's skew decays to 0.494 of its one-year value by three years, against the market's 0.644: too flat at the short end and decaying too fast at the long end simultaneously, which is not a parameter problem but a structural one. Fitting Heston to a whole surface therefore means fitting it badly everywhere rather than well somewhere.

Jumps fix the short end and overshoot it, as Example 21.3 showed: ratio 3.7 against the market's 2.7, the third curve in Figure 21.4. Bates, the fourth curve, splits the difference at the short end (2.92 against 2.70) and inherits the jump model's long end, 0.372 against 0.644, for the reason Section 21.6 gave.

None of the three produces the power law. In each of them the variance is either constant or a Markov diffusion, with or without jumps appended to the price, and the short-end skew of any such model either saturates or is set by hand, maturity by maturity, through the jump parameters. The market's curve is not a compromise between the other three; it is a different shape, and matching it needs either a different kind of process or several ordinary ones with widely separated speeds. That was the position for two decades: a short-maturity problem, patched and never explained.

21.8 ROUGH VOLATILITY

In a paper circulated in 2014 and published in 2018, Gatheral, Jaisson, and Rosenbaum estimated the smoothness of realized volatility directly from high-frequency data on equity indices and liquid futures, and found a pattern that had been in the data all along.

Log-volatility behaves like the fractional Brownian motion of Section 14.13, with Hurst parameter $H \approx 0.1$. Ordinary Brownian motion has $H = 0.5$; at 0.1 the path is far *rougher* (more jagged, with successive increments correlated at -0.43 rather than zero), and the rough path of Figure 14.5 is what such a series looks like. The estimate was strikingly stable: roughly

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the same H across twenty-one equity indices and for Bund, crude oil and gold futures, between 0.06 and 0.20, and nowhere near 0.5, when the sample was split in two, and, in later work, across close to two thousand individual US stocks.

Two things made this more than an empirical curiosity.

It produces the observed skew term structure with no extra machinery. A model whose volatility is driven by a fractional Brownian motion with Hurst parameter H generates

$$\psi(T) \sim T^{H-1/2} \quad \text{as } T \to 0 \tag{21.8}$$

where $\sim$ means proportional to, in that limit. Setting $H = 0.1$ gives $\psi(T) \sim T^{-0.4}$—the observed power law, from one parameter, with $H = 0.5$ recovering the flat short-end behavior of the classical models. Figure 21.4's "market" curve *is* equation (21.8) at $H = 0.1$.

And roughly the same H appears on both sides. The value estimated from realized volatility under the physical measure (the real-world probabilities that generated the data) roughly matches the value implied by option prices under the risk-neutral measure. A parameter estimated from time-series data has no obligation to match one estimated from cross-sectional prices, and when it does, that is evidence about the process rather than about the fit.

Case Study 21.1 How roughness became usable

Rough volatility was, at first, a model no one could use. Fractional Brownian motion is not Markovian: the expected future path of volatility depends on its entire history, so there is no state variable to put on a tree and no partial differential equation to solve. Every price required a simulation, and simulating the model's volatility exactly, jointly with the price, costs $O(n^2)$ per path in the number of time steps after a Cholesky factorization, against $O(n)$ for an ordinary diffusion.

Three developments changed that between roughly 2015 and 2021.

Fast simulation. Hybrid schemes reduced the cost of generating rough paths to something comparable with ordinary Monte Carlo, by computing exactly the heavily weighted most recent shocks and approximating the weight on older ones.

Markovian approximations. The rough weighting of past shocks can be approximated by a sum of a small number of ordinary mean-reverting factors with widely separated speeds. This restores a state vector, typically of four to eight dimensions, and with it ordinary Monte Carlo and, in low dimension, the partial differential equations of Chapter 20. It is an approximation, but a controlled one, and it is how most implementations work.

Neural network calibration. Calibrating a rough model by repeated simulation is prohibitive. Training a network offline to map parameters to implied volatility surfaces, then inverting it at run time, turns a calibration that took hours into one that takes milliseconds. The network forecasts nothing; it interpolates a known function that is merely expensive to evaluate, so it replaces repeated simulation without changing the model being calibrated.

Adoption has been uneven. Non-bank market makers and some hedge funds took the models up first, for short-dated index options and relative-value trades between SPX and VIX options; banks have been slower, and most exotics books remain on the local-stochastic volatility of Section 21.9.

The lesson is about what a modeling constraint really is. For twenty years the profession assumed volatility must be modeled as a Markov diffusion, and most failures at the short end were patched by adding jumps. The Markov assumption was never an economic claim; it was a computational one, and when the computation became affordable the assumption went. It is worth asking which of today's modeling conventions are economics and which are just arithmetic we have not yet made cheap enough.

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The standard implementation is the **rough Bergomi** model of Bayer, Friz, and Gatheral (2016), and it is worth writing out, because each of its three parameters has one main job:

$$\begin{aligned} \frac{dS}{S} &= (r-q)\,dt + \sqrt{v_t}\,dW_{1,t}, \qquad dW_1\,dW_2 = \rho\,dt \\ v_t &= \xi_0(t)\exp\left\{\eta\sqrt{2H}\int_0^t (t-s)^{H-1/2}\,dW_{2,s} - \frac{\eta^2}{2}\,t^{2H}\right\} \end{aligned} \tag{21.9}$$

Read it from the outside in. $\xi_0(t)$ is the **forward variance curve**, today's price of the variance to be realized at t, which follows from the variance swap rates Section 19.3 replicated from the vanilla surface and which Chapter 22 trades. It is an input rather than a parameter: the exponential has mean one, so $\mathbb{E}[v_t] = \xi_0(t)$ and the curve is reproduced exactly whatever the other three do. The integral is a version of Section 14.13's fractional Brownian motion (the Riemann–Liouville form: the same roughness H, but increments that are not stationary), written as a weighted sum of past shocks: each $dW_{2,s}$ enters with weight $(t-s)^{H-1/2}$, the **kernel** of Section 14.13 (Heston's is $e^{-\kappa(t-s)}$), and the $\sqrt{2H}$ normalizes the variance to t^{2H}, which is why $-\frac{1}{2}\eta^2 t^{2H}$ keeps the mean of the exponential at one. η scales the volatility of log variance and ρ the correlation, with the same jobs as Heston's σ_v and ρ.

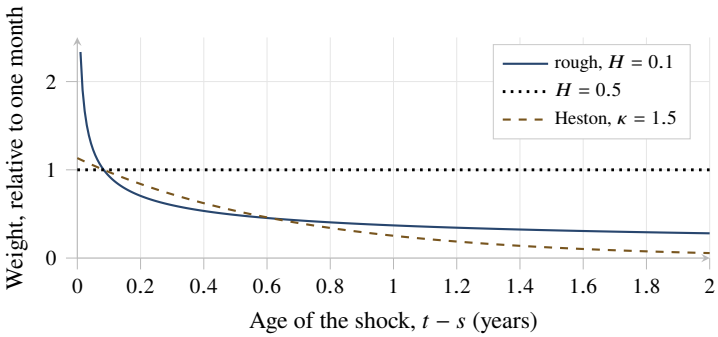


Figure 21.5: Weight on a past variance shock as a function of its age, normalized to one at one month.

The kernel is where the model differs from everything before it, and Figure 21.5 draws it against Heston's. For $H < 1/2$ the exponent is negative and the weight rises without limit as $s \to t$: the most recent shocks dominate, which is roughness. But the weight decays as a power rather than an exponential, so a shock from two years ago still carries 28% of the weight of one from a month ago, where Heston's $e^{-\kappa(t-s)}$ at $\kappa = 1.5$ leaves it 6%. Rough volatility is both more jagged and longer-remembering than a mean-reverting diffusion, and the combination is what a single exponential cannot produce: to move that much over short horizons, Heston needs a very large σ_v; to remember that long, a very small κ; and the two settings fight over the same one-year smile. Table 21.4 puts numbers on the three parameters in the format of Tables 21.2 and 21.3. Its figures are simulated (200,000 paths, hybrid scheme), so skews are reliable to about ±0.01 and the ratio to about ±0.05; skew and curvature are central differences at a half-width $h = 0.10$, and the ratio uses the ±5% strike pair, since Monte Carlo noise rules out Table 21.3's $h = 0.01$, so it is not the quotient of the skew columns either.

Two of the three read exactly as they did for Heston. *ρ sets the tilt*: at $\rho = 0$ the skew is zero at both maturities and the smile is symmetric. *η sets the curvature*, and drags the level with it as σ_v did: from $\eta = 1.0$ to 3.0 the one-month curvature goes from −0.1 to 13.3 and the at-the-money level from 19.3% to 15.2%.

The third has no counterpart in Heston. From $H = 0.10$ to 0.45 the one-month/one-year skew ratio falls from 2.6 to 1.3, where (21.8) predicts $12^{1/2-H}$: 2.70, 1.86 and 1.13, which is

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Table 21.4: One-month and one-year at-the-money level, skew, and curvature as each parameter is varied from the rough Bergomi base case ($H = 0.10$, $\eta = 1.9$, $\rho = -0.9$, flat $\xi_0 = 0.04$).

	One month			One year			
	ATM (%)	Skew	Curv.	ATM (%)	Skew	Curv.	**Ratio**
Base	17.8	−0.63	4.3	16.3	−0.31	0.0	2.6
$H = 0.25$	18.6	−0.73	0.1	15.1	−0.39	−0.3	2.0
$H = 0.45$	19.5	−0.47	−0.3	15.3	−0.36	−0.2	1.3
$\eta = 1.0$	19.3	−0.51	−0.1	18.6	−0.19	−0.1	2.7
$\eta = 3.0$	15.2	−0.47	13.3	12.8	−0.30	2.1	2.1
$\rho = -0.5$	18.3	−0.34	5.9	17.1	−0.17	0.9	2.4
$\rho = 0$	18.6	0.00	6.7	17.7	0.00	1.2	—

close, given that (21.8) describes the skew as maturity shrinks and is only approximate between a month and a year (Heston, a diffusive-variance model with $H = 1/2$, gives 1.77, not 1). The one-year skew, meanwhile, stays between −0.31 and −0.39 across the same three rows. H is a term-structure parameter: it decides how the skew is distributed across maturities, not how much of it there is at a year, and Heston has nothing that does that job. Two further points attach to the table. The parameters interact: the standard deviation of log variance over a horizon t is $\eta\, t^H$, so with the same η a rougher process swings more within a month, and the one-month level falls as H falls; over a year those swings largely average out, so the rough row's average variance is the least dispersed and its one-year level the highest. And on index surfaces three parameters, with the forward variance curve taken from the market, fit better than Heston's five (Bayer, Friz, and Gatheral, 2016), which is the strongest argument for the model.

Estimating H. The number did not come from option prices. Gatheral, Jaisson, and Rosenbaum built a daily series of realized volatility, σ_t on day t being the square root of that day's realized variance from five-minute returns, took its logarithm, and for each lag Δ from one to about thirty days computed the average p-th power of the change over that lag,

$$m(p, \Delta) = \frac{1}{n} \sum_{t} \left| \log \sigma_{t+\Delta} - \log \sigma_t \right|^p$$

over the n pairs of days in the sample, for powers p from about 0.5 to 3. Averaging over start days measures a single quantity because the processes they fitted, fractional Brownian motion and its slowly mean-reverting version, have stationary increments: a change over Δ days has the same distribution whichever day it starts on. Self-similarity then gives $m(p, \Delta) \propto \Delta^{pH}$, so a regression of $\log m$ on $\log \Delta$ has slope pH; the slopes came back proportional to p, as the model requires, with H of order 0.1, and nowhere near 0.5, for every index examined. Two cautions attach to the number. Realized variance is an estimate of the variance, not the variance itself, and measurement noise in it makes a series look rougher than it is; how much of the measured roughness is noise is still debated, as the end of this section explains. And the estimate is made under the physical measure; the H that prices options is read off the skew term structure under the risk-neutral measure, and the two roughly agreeing, as the start of this section noted, is the evidence, not an assumption.

Where the frontier is. Two developments since 2019 matter. Rough Heston replaces the exponential decay of a variance shock in Heston (Figure 21.5) with the power-law kernel. Because the model is still affine (the logarithm of its characteristic function is linear in today's forward variance curve, as Heston's is in v_0 above), its characteristic function survives: it

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solves a fractional Riccati equation numerically rather than in closed form, but one integral still prices a vanilla and the machinery of Section 21.4 carries over. And the joint calibration problem, one model fitting the index smile and the VIX smile of Chapter 22 at the same time, long defeated the classical models and was narrowed but not closed by rough volatility. Among the models that fit both, path-dependent volatility makes today's volatility a weighted function of the index's own past path, its recent returns and their recent squares, rather than of a separate factor. That is Section 19.9's leverage effect, seen from the modeling side.

Two caveats follow. Rough models fit vanilla surfaces very well and their advantage for pricing exotics is real but smaller than the vanilla fit suggests. And the choice of H is not innocuous for anything sensitive to the very short end, where the model's predictions are most distinctive and market data are thinnest.

What is still contested. Each leg of the case for roughness has been questioned since 2022. On S&P 500, Euro Stoxx 50 and DAX surfaces, El Amrani and Guyon (2023) find that the power law fits from about a month to three years but not below a month, where a skew rising toward a finite limit fits better. Cont and Das (2024) show that realized volatility estimated from a process that is not rough can still measure as rough, so the time-series estimate of H may be partly an artifact of estimation error. And the quintic Ornstein–Uhlenbeck model (a polynomial in one mean-reverting Gaussian factor) of Abi Jaber, Illand, and Li (2023) fits SPX and VIX smiles jointly with a single Markov factor and no rough paths, its steep short-end skew coming from fast mean reversion and a large vol-of-vol. Rough volatility remains the most economical account of the skew term structure; whether volatility is in fact rough is open.

21.9 LOCAL-STOCHASTIC VOLATILITY

Desks divide the work. Local volatility serves for payoffs the vanilla surface alone determines, and jump models mainly the short-dated skew; for path-dependent equity and foreign-exchange exotics, what desks run is neither the model of Section 21.2 nor that of Section 21.4. It is both:

$$dS = (r - q)S\,dt + L(S,t)\sqrt{v}\,S\,dW_1 \tag{21.10}$$

with v following a stochastic-volatility process as in (21.4). The function $L(S,t)$, the **leverage function**, is chosen so that the model reproduces the vanilla surface exactly.

The stochastic part supplies realistic dynamics; the leverage function absorbs whatever is left over so the fit is exact. What follows says what L is, what the one free choice in the construction does to a price, and who gets to make that choice.

What the leverage function is. A model reproduces the vanilla surface if and only if the average of its instantaneous variance, conditional on the spot at each date, equals the Dupire local variance at that spot, the result [TN21] derives from Gyöngy's theorem; local variance is therefore itself an average, of the instantaneous variance over the paths that pass through (S,t). For (21.10) the instantaneous variance is $L^2 v$, so

$$L^2(S,t) = \frac{\sigma^2_{\text{loc}}(S,t)}{\mathbb{E}\left[\, v_t \mid S_t = S \,\right]} \tag{21.11}$$

The numerator is the variance the surface demands at (S,t); the denominator is the variance the stochastic factor delivers there on average. The leverage function is the part of the smile the stochastic factor did not explain: above one where the stochastic model under-delivers, below one where it over-delivers, and exactly one where it already fits. The equation is circular, since the conditional expectation on the right depends on the paths and the paths depend on L. It is solved forward in time, one step at a time, by a forward partial differential equation or

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by a particle method that estimates the expectation from the simulated paths as they advance. [TN21] sets out the fixed point and both ways of solving it.

The mixing parameter. Scale the volatility of volatility, $\sigma_v \to \zeta\,\sigma_v$ with $\zeta \in [0, 1]$, and recalibrate L at each setting. At $\zeta = 0$ the variance is deterministic, the denominator of (21.11) is just $v(t)$, and $L = \sigma_{\text{loc}}/\sqrt{v}$: pure local volatility. At $\zeta = 1$ the stochastic factor explains as much of the smile as it can and L carries only the residual. Every setting reproduces the vanillas exactly, which is what makes ζ different from every other parameter in this chapter: it is not a *fit* parameter. It is a choice about dynamics that the vanilla market cannot make for you, and Figure 21.6 shows what it does to L. The vanilla surface there is the whole implied-volatility surface, every strike and maturity, that Heston produces at the base case of Figure 21.2, and the stochastic factor is that same base-case variance process, so at $\zeta = 1$ the stochastic model already fits and L equals one everywhere—the check that the construction is working. At $\zeta = 0$ the leverage function is the entire local-volatility shape, 1.51 at a spot of 75 falling to 0.57 at 115, and at $\zeta = 0.5$ it is flatter but not a midpoint: just below the money it dips below both, because the half-strength stochastic factor delivers more at-the-money variance than the surface wants.

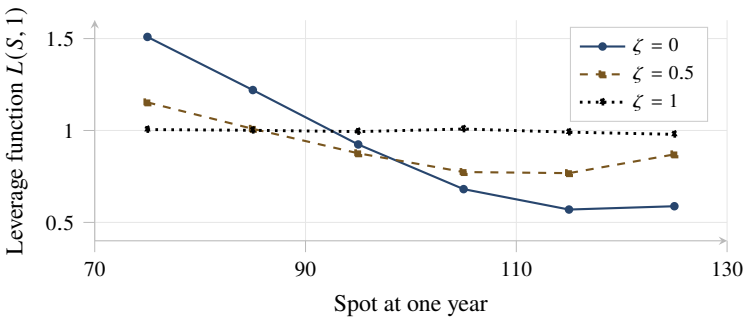


Figure 21.6: The leverage function at one year on Heston's own surface, at three settings of the mixing parameter.

Example 21.5 The mixing parameter as a dial

Heston's base case generates the vanilla surface; the local-stochastic model (21.10) is calibrated to it at five settings of ζ by the particle method, and prices a one-year up-and-out call, strike 100, barrier 110, monitored daily (extinguished when the spot first reaches 110); P(survive) is the share of paths that never touch the barrier. The particle method runs 80,000 Monte Carlo paths; the vanilla check is the one-year implied volatility the model returns at three strikes, against Heston's 23.37%, 16.94% and 13.33%.

	One-year vanilla fit				
Mixing ζ	$K = 80$	$K = 100$	$K = 120$	**Up-and-out 110**	**P(survive)**
0	23.35%	16.96%	13.30%	0.420	44.8%
0.25	23.34%	16.94%	13.29%	0.463	45.7%
0.5	23.27%	16.89%	13.28%	0.572	47.7%
0.75	23.21%	16.85%	13.23%	0.707	50.2%
1	23.24%	16.85%	13.23%	0.845	52.7%

The vanilla columns are flat. Every row matches the target to within 0.2 volatility points, which is the noise of the simulation and of averaging within spot ranges: the fit is exact at every setting, as the construction promises. *The barrier column doubles.* From $\zeta = 0$ to 1 the price of the same

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option, against the same vanilla surface, goes from 0.420 to 0.845, and it moves monotonically, so any value between the two can be produced by a setting of the dial. Both ends are legitimate calibrated models. Nothing the vanilla market quotes distinguishes them. The two ends are the two models of Example 21.7, whose independent simulation prices the same option at 0.4244 and 0.8402; the differences from 0.420 and 0.845 here are within the two runs' standard errors (about 0.005 and 0.008).

Section 21.11 returns to this option and explains the direction: a stochastic-volatility path that ends between the strike and the barrier is far more likely to have stayed below the barrier.

That is why the mixing parameter is not a nuisance. *It is the model risk, made into a dial*: a desk that prices a barrier at both ends of it has measured its own exposure to the modeling choice without building a second model, and the range (here a factor of two) is the number to take to a risk committee, not either end of it.

Who sets the dial, and how. The model is the standard on two kinds of desk, and they set ζ differently. In foreign exchange, where local-stochastic volatility has been the standard model for barrier and touch products since the mid-2000s, one-touch and double-no-touch options (which pay a fixed amount if the exchange rate touches a level, or if it never leaves a range, before expiration) trade actively enough to calibrate to: the desk picks the ζ that reproduces their prices, and the market has, in effect, priced the dial. In equities there is no such instrument. Barriers, autocallables and cliquets trade over the counter and are priced by the same desks that would need them as inputs, so ζ is set by judgment (from the historical volatility of implied volatility, from how the forward smile has behaved, or by house policy), and Section 21.11 will argue that the honest output is then the range rather than a point. The products most sensitive to the setting are the ones whose value depends on the forward smile: autocallables and the snowballs of Chapter 35, whose knock-in put has its barrier where the put is in the money, the reverse-barrier case on which Example 21.7 shows the models disagree.

Living with it. Two practical consequences follow from (21.11). Because L is a function of today's surface, it is not calibrated once and kept; it is re-derived whenever the surface moves, which in practice means daily, and the recalibration is the expensive part of running the model. And because the denominator is a conditional expectation estimated from finitely many paths, L is noisiest in the wings, where paths are few. A leverage function that looks jagged there is an estimator artifact, not a discovery, and practitioners smooth it before use. The delta and vega the model produces lie between local volatility's and Heston's, so the hedge, not only the price, moves with ζ.

21.10 CALIBRATION IN PRACTICE

Every model above has to be fitted before it can be used, and the fitting is where most of the practical damage is done. Three decisions matter more than the choice of model.

What to fit to. A surface contains hundreds of quotes of wildly different quality: the at-the-money one-month option trades continuously, the five-year 10-delta put trades twice a month. Fitting to all of them equally lets the least reliable quotes drive the parameters.

The standard remedies are to fit implied volatilities rather than prices (equivalently, to divide each price error by the option's vega) and to weight each quote by its liquidity, through bid–ask width or open interest. Dividing by vega is the most common choice, because it measures every error in volatility points, the unit in which options are quoted and bid–ask spreads are set, so a wing option's error counts as much as an at-the-money one. Weighting volatility errors *by*

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vega does the opposite, turning them back into price errors, which the at-the-money options dominate.

Fit quality against parameter stability. A model refitted every day to the whole surface will produce parameters that jump. Some of that is real; most of it is the optimizer trading one parameter against another along the flat directions of the objective function, which the overlapping effects in Table 21.2 create.

The consequence is not cosmetic. A book revalued with jumping parameters generates phantom profit and loss, and a desk cannot distinguish a genuine mark-to-market move from a calibration artifact. Standard practice is therefore to penalize movement away from the previous day's parameters (a regularization term), accepting a slightly worse fit in exchange for parameters that mean something over time.

Which risk the calibration is for. A model fitted to the whole surface is fitted to no part of it especially well. If the book's exposure is concentrated in one-year 25-delta puts, that is what the calibration should target, and the fit elsewhere matters little.

The objective should match the hedge: a model that reproduces every quote and mishedges the one position on the book has failed. Example 21.6 shows the objective alone moving the fit.

Example 21.6 Two fits, one surface

Fit Heston to the jump model's surface of Example 21.3, which no Heston parameters reproduce, at one month, three months and one year, strikes 80 to 120. One fit minimizes squared price errors; the other first divides each price error by the option's vega, which to first order makes it an implied-volatility error.

Price errors are dominated by at-the-money options, which have the largest vega, so a given volatility miss costs most there. The one-month 80 put is worth \$0.05, so the first fit leaves its implied volatility 7.08 points too low while missing the one-month at-the-money option by 0.12 points. The second cuts the wing miss to 4.01 points at a cost of 0.19 points at the money, and the average absolute one-month miss falls from 1.72 points to 1.11. Weighting volatility errors *by* vega instead nearly gives back the first fit (6.67 points in the wing).

Neither is right in general. The first is right for a desk whose book is at-the-money straddles; the second for a desk whose book is in the wings, like the structured-product issuers of Section 19.8. The objective encodes an assumption about the book; choose it rather than inherit a software default.

21.11 MODEL RISK MADE CONCRETE

Section 19.12 asserted that two models fitting today's surface perfectly can price the same barrier option far apart. That assertion is now checkable, because Sections 21.2 and 21.4 give two models that agree on every vanilla by construction.

Example 21.7 The same vanillas, two barrier prices

Take the Heston parameters of Figure 21.2, and derive the local-volatility surface from Heston's own vanilla prices by (21.2). The two models now agree on *every* European option: priced by simulation, the one-year 100-strike call comes to 6.766 under Heston and 6.767 under local volatility, against the semi-analytic 6.751, equal within Monte Carlo error.

Now price barrier options under each, one year, monitored daily, using the *same random numbers* for both models so that the difference is measured directly rather than as the difference of two noisy prices. This is Section 20.10's common-random-numbers technique.

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Barrier	Heston	Local vol	Difference (± s.e.)	Relative
Down-and-out call, strike 100				
80	6.5286	6.5821	+0.053 ± 0.015	+0.8%
85	6.2180	6.2962	+0.078 ± 0.015	+1.3%
90	5.5388	5.6138	+0.075 ± 0.015	+1.4%
Up-and-out call, strike 100				
130	5.8138	5.5787	−0.235 ± 0.015	−4.0%
120	4.0046	3.4193	−0.585 ± 0.013	−14.6%
115	2.4071	1.7138	−0.693 ± 0.010	−28.8%
110	0.8402	0.4244	−0.416 ± 0.005	**−49.5**%

On the down-and-out the two models agree to within about one percent, a difference a desk could ignore. On the up-and-out they disagree by nearly half the price.

So model risk is not a property of "barrier options." It is a property of where the payoff lives relative to where the models disagree. A down-and-out call struck at the money pays off in a region well above the barrier where the two models describe the distribution almost identically, because the vanillas pin it down. A reverse knock-out (the case Section 19.12 described, as this up-and-out call is) pays off only in the narrow band between the strike and the barrier, and that band is exactly where the surface constrains the models least: no vanilla option isolates the joint behavior of "finish in this band" and "never touch the top of it."

The direction is worth stating carefully, because a plausible-sounding argument gets it backward. One might reason that local volatility is low at high spot (Example 21.1 gives 12.0% at a spot of 110 a year out, against 16.2% at 100), so paths approach the upper barrier slowly, knock out less, and the option should be worth *more*. It is worth less, and the premise is false: local-volatility paths knock out more often (44.8% survive against 52.7% in Example 21.5). Both models put the same probability on every terminal level, so what matters is survival *given* where a path ends. Of the paths finishing between 108 and 110, 39% have never touched 110 under Heston and 14% under local volatility; of those finishing between 100 and 102, 77% against 57%. Under Heston some paths are quiet and some wild, and the quiet ones reach the band without touching the barrier; local volatility reproduces the same terminal distribution with one volatility at each spot and date, so it has no quiet paths to offer.

So the vanilla market does not price this product, no recalibration will make it, and the number to report is the range the mixing parameter of Example 21.5 produces, which matters only where it is wide; this table says which products those are, before the trade.

Case Study 21.2 When the model choice showed up in the profit and loss

Autocallables are the clearest case of this chapter's model choice paid for in cash.

The structure, described in Section 12.12 and Chapter 35, embeds a down-and-in put that the investor sells to the issuer and a series of discrete observation dates. Its value depends on three things the vanilla surface does not determine: how the smile evolves between observation dates, volatility near the barrier, and, for notes on several underlyings, how their correlation rises in a decline.

Issuers hedge these books dynamically, and the hedge is computed from a model. Through the 2010s many issuers priced them on local volatility, because it fitted the vanillas exactly and the fit was auditable. Example 21.2 identifies the flaw in that assumption. The issuer is long the knock-in put but short the coupon and early-redemption payments, which are digitals, and a steeper forward skew raises the value of both, so the sign of the net exposure depends on the barriers and the coupon. On many notes the digitals dominate, the issuer loses when forward skew steepens, and local volatility, which predicts

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a flattening forward smile, *understates* what the issuer owes; on any note it puts the vega profile in the wrong maturities.

The consequence stays hidden while the market is calm and appears when the index falls toward the knock-in region, when the forward smile matters most and the misspecified hedge has to be adjusted quickly. Section 18.12 described the resulting flow (issuers buying index futures into the decline, then selling, and withdrawing that bid, at the barrier); the model question is why the hedge needed adjusting then rather than earlier.

Desks moved to local-stochastic volatility for these books, and the mixing parameter of Section 21.9 is now a first-order risk number rather than a technical setting. They moved because the simpler model's error had a price, discovered in a drawdown.

The general lesson is that a model's failure mode determines what it may price: local volatility fits and mis-forecasts, stochastic volatility forecasts and mis-fits, and products that depend on both need the two combined, with the remaining uncertainty in the combination's free parameter.

21.12 CHOOSING A MODEL

The decision is not about realism. It is about which failure the trade is exposed to.

Marking a vanilla book → no model at all: interpolate the surface of Section 19.12.

Pricing a barrier or an autocallable → local-stochastic volatility, and report the range across the mixing parameter, since here the choice has a material cost.

Forward-starting products, cliquets → stochastic or rough volatility. Local volatility gets the forward smile structurally wrong (Example 21.2), and the forward smile is the product.

Very short-dated options → a smile fitted directly on a business-time clock (Section 23.9), with rough volatility or jumps as the check that some process could produce it. Classical stochastic volatility cannot make the skew, and Chapter 23's 0DTE market lives entirely in this region.

Swaptions and rate smiles → SABR, which Section 29.9 sets out and calibrates.

Two habits follow, parallel to Chapter 20's.

Calibrate to what you will hedge with. Fit to the instruments that will actually be in the hedge, and check the fit there rather than in aggregate.

Price it twice, with different models. The spread is not noise to be averaged away; it is the number the trade's model risk is measured in, and Example 21.7 shows how far two models can differ while agreeing on everything observable.

21.13 WHERE THIS CAME FROM

The Black–Scholes–Merton formula was accurate enough that its failures took more than a decade to become commercially serious, and the crash of 1987 made them so. The equity smile appeared and did not go away.

Three answers were soon available, two of them already in print before the crash, and all three survive. Merton (1976) had already added jumps, which produce a smile at short maturities and were the natural response to a market that had just gapped. Hull and White (1987), with Wiggins (1987) and Scott (1987) independently, made volatility itself stochastic (the standard reference, and still the paper to read for the structure of the problem). Stein and Stein (1991) added a tractable mean-reverting form, and Heston (1993) displaced them all—not because its dynamics are better but because its characteristic function is known in closed form and the price is therefore a single integral. Carr and Madan (1999) made that computational accident decisive by showing how to invert such transforms fast.

The other answer was to give up choosing the dynamics. Dupire (1994), with Derman

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and Kani (1994) and Rubinstein (1994) arriving from trees, showed that exactly one local volatility surface reproduces every quoted vanilla. This chapter is largely about the price of that exactness. Bates (1996) combined jumps with stochastic volatility, the form in which jump models are now mostly used. Hagan and co-authors (2002) then gave SABR, whose asymptotic formula made a stochastic-volatility smile quotable from a formula and became the market's language for rates. And Gatheral, Jaisson, and Rosenbaum (2018) reopened the question by measuring that volatility is rougher than any of these assume.

Section 21.11's method, reading the spread between models fitted to one vanilla surface as the model risk, is Hull and Suo's (2002). They applied it to the implied volatility function model (local volatility) and found what Section 21.11 finds: agreement on every quoted vanilla leaves some exotics nearly fixed and barrier options far apart. The contribution was to make model risk a measurement rather than a warning, and it recurs in Chapters 34 and 39.

SUMMARY

A volatility model must fit today's surface and predict how it moves. Nothing in the vanilla market pins down the second requirement, and that degree of freedom is what every model in this chapter spends differently.

Local volatility makes σ a deterministic function of price and time, read directly off the surface by Dupire's formula, so the fit is exact by construction and the market stays complete. The local volatility is about twice as steep in strike as the implied volatility, because an implied volatility is an average of the local volatilities a path visits, and averaging flattens. Its failure is dynamic: the local-volatility skew decays steeply with maturity, so the model predicts a nearly flat forward smile where the market's is roughly stationary, and forward-starting products are mispriced and hedged with the wrong vega.

Heston makes variance a mean-reverting square-root diffusion correlated with the price. Correlation produces the skew, vol-of-vol produces the curvature, and mean reversion flattens both. No parameter moves one feature alone, so calibration is a genuine optimization and individual fitted parameters are not separately meaningful. Jumps produce short-dated skew no diffusion can, but that skew collapses with maturity faster than the market's does.

The at-the-money skew follows a power law $\psi(T) \sim T^{-0.4}$ from about a month to a few years; below a month the evidence is disputed. Heston saturates instead, rising too little toward the short end, and decays too fast at the long end: missing both ends at once is a structural failure, not a parameter problem. Bates, which runs a variance diffusion and jumps together, brings the short end close to the market's and pays for it at the long end. Eight parameters place the fit; they do not produce the law.

Rough volatility is the most economical resolution, though each leg of its case is now contested. Log-volatility behaves like a fractional Brownian motion with $H \approx 0.1$, which yields $\psi(T) \sim T^{H-1/2} = T^{-0.4}$ from a single parameter, and roughly the same H appears in realized volatility and in option prices. Rough Bergomi, given the market's forward variance curve, needs three parameters and fits index surfaces better than Heston. What made it usable was fast simulation, Markovian approximations that restore a state vector, and neural-network calibration: the Markov assumption it displaced was a computational convention, never an economic claim.

Exotics desks run local-stochastic volatility, a stochastic variance process multiplied by a leverage function calibrated to make the vanilla fit exact, and the mixing parameter between the two is the model risk turned into a dial. Two models agreeing on every vanilla can still price the same barrier far apart, because they differ in how the price travels to each terminal distribution, so the output is a range, not a price. Calibration is a modeling choice too: the objective decides which quotes the fit honors and should match the book's hedge, and parameters that stay stable

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from day to day are bought with a worse fit.

FURTHER READING

Gatheral's *The Volatility Surface* treats local and stochastic volatility together and is the natural next step from this chapter. Dupire (1994) is short and worth reading in the original for the argument rather than the formula. On Heston, the original 1993 paper sets out the model, but the numerically important material (branch cuts in the characteristic function, and the "little trap" formulation that avoids them) is in Albrecher and co-authors and in Gatheral, and any implementation that skips it will produce silently wrong prices at long maturities. Hagan and co-authors (2002) is the SABR paper; the wing corrections that repair its negative densities are later and scattered. For rough volatility, Gatheral, Jaisson, and Rosenbaum (2018) is the empirical paper that started it and is explicit about what was measured; Bayer, Friz, and Gatheral (2016) introduce rough Bergomi, and El Euch and Rosenbaum (2019) give rough Heston its characteristic function, which is the paper to read for how the affine machinery survives roughness. Bates (1996) is the paper that adds jumps to stochastic volatility, and Duffie, Pan and Singleton (2000) the general affine framework that explains why its characteristic function factorizes. For the local-stochastic model, Guyon and Henry-Labordère's *Nonlinear Option Pricing* (2013) is the account of the particle method by the people who made it practical, and is clear that the difficulty is the conditional expectation rather than the calibration. Guyon and Lekeufack (2023) is the path-dependent volatility paper behind the frontier remark in Section 21.8. On calibration itself, Christoffersen and Jacobs (2004) is the paper to read before choosing an objective function: fitting the same model to dollar price errors and to implied-volatility errors produces different parameters and different out-of-sample performance, which is why Section 21.10's weighting question is a modeling decision rather than a numerical one. The literature on Markovian approximations and on neural-network calibration is recent and moving, and a reader should expect the state of the art to have advanced.

PRACTICE QUESTIONS

21.1 Explain why the denominator of Dupire's formula (21.2) is proportional to the implied density, and what that implies about when a local-volatility model exists.

21.2 Explain the "rule of two" using the averaging argument, and state whether a steep local-volatility surface is evidence of a calibration error.

21.3 Using Example 21.2, explain why a local-volatility model misprices a forward-starting option and in which direction.

21.4 A Heston fit returns $\rho = -0.05$. What shape of smile does this imply, and in which market would you expect to see it?

21.5 Explain why raising σ_v changes the at-the-money level as well as the curvature, and what this implies about interpreting a single fitted parameter.

21.6 State what jumps produce that a diffusion cannot, and state the corresponding defect in the term structure they generate.

21.7 Define the at-the-money skew $\psi(T)$, state the empirical power law it obeys, and explain why a classical stochastic-volatility model cannot reproduce a skew that keeps steepening as maturity shortens.

21.8 State the relationship between the Hurst parameter and the skew term structure, and give the value of H that reproduces the observed exponent.

21.9 Explain what the leverage function in a local-stochastic volatility model does, and what

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the mixing parameter measures.

21.10 Two models reproduce every quoted vanilla price and disagree on a barrier. Explain how this is possible without either committing an arbitrage, and state what a desk should report as a result.

21.11 Explain why a calibration objective is usually stated in implied volatility (price errors divided by vega) rather than in price, what a price-error objective would be blind to, and what a desk gains and gives up by penalizing day-to-day parameter movement.

FURTHER QUESTIONS

21.12 Example 21.1 finds a local-to-implied slope ratio of 1.75 rather than exactly 2. Derive the averaging argument that produces the factor of two, identify the approximation under which it is exact, and explain what features of the surface make the realized ratio depart from it.

21.13 Section 21.3 argues that local volatility gets the forward smile wrong while fitting today's surface exactly. Construct two processes that produce identical terminal distributions at every maturity but different forward smiles, and use them to explain precisely what information about the process the vanilla surface does not contain.

21.14 Table 21.2 shows that a rise in κ and a fall in σ_v both flatten the smile. Show that they do so through different mechanisms, explain why this makes them hard to identify jointly from one maturity, and describe what data would separate them.

21.15 Section 21.7 shows Heston too flat at the short end and decaying too fast at the long end simultaneously. Explain why adding parameters cannot fix both, and state what this implies about the practice of calibrating a single model to an entire surface.

21.16 Equation (21.8) gives $\psi(T) \sim T^{H-1/2}$. Explain why $H < 1/2$ makes the skew diverge as $T \to 0$, discuss whether an infinite short-maturity skew is economically sensible, and state what the model is really claiming about very short-dated options.

21.17 The case study in Section 21.8 argues that the Markov assumption in volatility modeling was a computational convention rather than an economic claim. Identify two other modeling conventions in this book that may be of the same kind, state what would have to become cheap for each to be abandoned, and explain how one would tell the difference between a convention and a claim.

21.18 Table 21.3 holds the variance rate fixed while the jump parameters change, and the one-year skew barely moves while the one-month skew moves a great deal. Using the fact that a jump component's contribution to skewness decays like $T^{-1/2}$ while a correlated diffusion's grows with maturity at first, explain why, and state what this implies about which options a Bates calibration should be run on to pin down the jump parameters.

21.19 Example 21.5 shows the up-and-out price rising monotonically with the mixing parameter while the vanilla fit is unchanged. Explain the mechanism, state why a foreign-exchange desk can calibrate the mixing parameter from market prices while an equity desk cannot, and describe what an equity desk should report instead of a single price.

21.20 Example 21.7 produces two barrier prices from models that agree on every vanilla. Explain why the local-volatility price is the lower of the two, construct a portfolio of vanillas that would partially hedge the difference, and explain why no portfolio of vanillas can hedge it completely.

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